JEE Main Online Exam 2020

Question with Solutions

3nd September 2020 | Shift-I

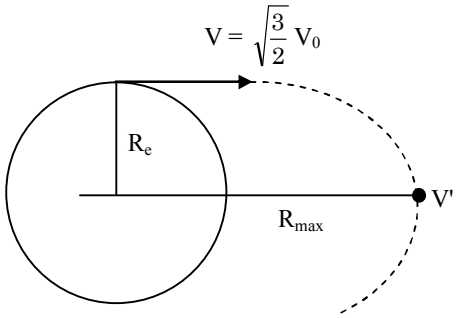
PHYSICS

Q.1 A satellite is moving in a low nearly circular orbit around the earth. Its radius is roughly equal to that of the earth's radius R_e. by firing rockets attached to it, its speed is instantaneously increased in the direction of its motion so that is become $\sqrt{\frac{3}{2}}$ times larger. Due to this the farthest distance from the centre of the earth that the satellite reaches is R. Value of R is

(1) $2R_e$ (2) $2.5R_e$ (3) $3R_e$ (4) $4R_e$

Ans. **[3]**

Sol.



$V_0 = \sqrt{\frac{GM}{R_e}}$

$\frac{-GMn}{R_e} + \frac{1}{2}mv^2 = \frac{-GMn}{R_{max}} + \frac{1}{2}mv'^2$ .....(i)

$VR_e = V'R_{max}$

Solving (i) & (ii)

$R_{max} = 3R_e$

Q.2 A balloon filled with helium (32°C and 1.7 atm) bursts. Immediately afterwards the expansion of helium can be considered as

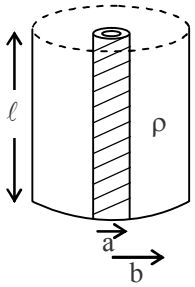
(1) irreversible adiabatic (2) reversible adiabatic

(3) irreversible isothermal (4) reversible isothermal

Ans. **[1]**

Sol. Bursting of helium balloon is irreversible & adiabatic.

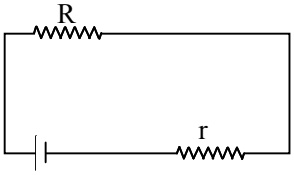
Q.3 Model a torch battery of length ℓ to be made up of a thin cylindrical bar of radius 'a' and a concentric thin cylindrical shell of radius 'b' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R, the maximum Joule heating in R will take place for



(1) $R = \frac{2\rho}{\pi\ell}\ell n\left(\frac{b}{a}\right)$ (2) $R = \frac{\rho}{\pi\ell}\ell n\left(\frac{b}{a}\right)$ (3) $R = \frac{\rho}{2\pi\ell}\left(\frac{b}{a}\right)$ (4) $R = \frac{\rho}{2\pi\ell}\ell n\left(\frac{b}{a}\right)$

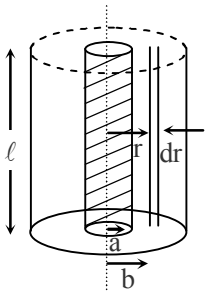
Ans. **[4]**

Sol. Maximum Power in external resistance is generated when it is equal to internal resistance of battery.



$P_R = \left(\frac{\in}{r+R}\right)^2 R$

P_R is max. when r = R



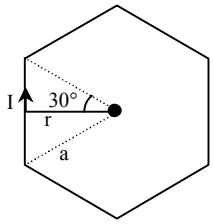
$$\int dr = \int_a^b \frac{\rho dr}{2\pi r\ell} \Rightarrow r = \frac{\rho}{2\pi\ell}\ell n\frac{b}{a}$$

Q.4 Magnitude of magnetic field (in SI units) at the centre of a hexagonal shape coil of side 10 cm, 50 turns and carrying current I (Ampere) in units of $\frac{\mu_0 I}{\pi}$ is-

(1) $5\sqrt{3}$ (2) $250\sqrt{3}$ (3) $50\sqrt{3}$ (4) $500\sqrt{3}$

Ans. **[4]**

Sol.



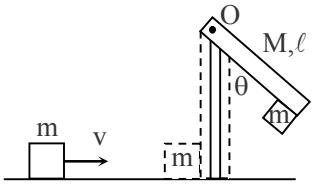
r = a cos 30°

$$B = \frac{6\mu_0 I}{4\pi a\cos 30^\circ} \times 2\sin 30^\circ \times 50$$

$$= \frac{\mu_0 I}{\pi}\frac{150}{\sqrt{3}a} = \frac{50\sqrt{3}}{0.1}\frac{\mu_0 I}{\pi}$$

$$= 500\sqrt{3}\ \frac{\mu_0 I}{\pi}$$

Q.5 A block of mass m = 1 kg slides with velocity v = 6 m/s on a frictionless horizontal surface and collides with a uniform vertical rod and sticks to it as shown. The rod is pivoted about O and swings as a result of the collision making angle θ before momentarily coming to rest. If the rod has mass M = 2 kg, and length ℓ = 1m, the value of θ is approximately (take g = 10 m/s^2)



(1) 63º (2) 69º (3) 49º (4) 55º

Ans. **[1]**

Sol. Angular momentum conservation

$$mv\ell = \frac{M\ell^2}{3}\omega + m\ell^2\omega$$

$$\Rightarrow \omega = \frac{1\times 6\times 1}{\frac{2}{3}+1} = \frac{18}{5}$$

Now using energy conservation

$$\frac{1}{2}\left(M\frac{\ell^2}{3}\right)\omega^2 + \frac{1}{2}(m\ell^2)\,\omega^2$$

$$= (m+M)r_{cm}\,(1-\cos\theta)$$

$$= (m+M)\left(\frac{m\ell + \frac{M\ell}{2}}{m+M}\right) g(1-\cos\theta)$$

$$\frac{5}{6}\times\left(\frac{18}{5}\right)^2 = 20(1-\cos\theta)$$

$$\Rightarrow 1-\cos\theta = \frac{18}{5}\times\frac{3}{20}$$

$$\cos\theta = 1 - \frac{27}{50}$$

$$\cos\theta = \frac{23}{50} \Rightarrow \theta \simeq 63^\circ$$

Q.6 When the wavelength of radiation of falling on a metal is changed from 500 nm to 200 nm, the maximum kinetic energy of the photoelectrons becomes three times larger. The work function of the metal is close to-

(1) 0.61 eV (2) 0.81 eV (3) 1.02 eV (4) 0.52 eV

Ans. **[1]**

Sol. $\frac{3}{1} = \frac{\frac{hc}{200\ nm} - \phi}{\frac{hc}{500\ nm} - \phi}$, hc = 1240 eV-nm

on solving $\phi = 0.61$ eV

Q.7 Two isolated conducting spheres S_1 and S_2 of radius $\frac{2}{3}R$ and $\frac{1}{3}R$ have 12 μC and –3 μC charges, respectively, and are at a large distance from each other. They are now connected by a conducting wire. A long time after this is done the charges on S_1 and S_2 are respectively.

(1) 6 μC and 3 μC (2) 4.5 μC on both

(3) 3 μC and 6 μC (4) +4.5 μC and –4.5 μC

Ans. **[1]**

Sol. Now

$Q_1 + Q_2 = Q'_1 + Q'_2 = 12\ \mu C - 3\ \mu C = 9\ \mu C$

& $V_1 = V_2 \Rightarrow \frac{KQ'_1}{\frac{2R}{3}} = \frac{KQ'_2}{\frac{R}{3}}$

$Q'_1 = 2Q'_2 \Rightarrow 2Q'_2 + Q'_2 = 9\mu C$

$\Rightarrow Q'_2 = 3\ \mu C$

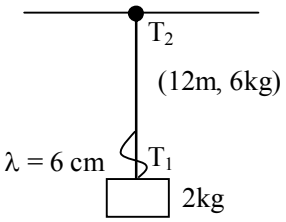
& $Q'_1 = 6\ \mu C$

Q.8 A uniform thin rope of length 12 m and mass 6 kg hangs vertically from a rigid support and a block of mass 2 kg is attached to its free end. A transverse short wavetrain of wavelength 6 cm is produced at the lower end of the rope. What is the wavelength of the wavetrain (in cm) when it reaches the top of the rope ?

(1) 12 (2) 9 (3) 3 (4) 6

Ans. **[1]**

Sol.



$V \propto \lambda$ $T_2 = 8g$

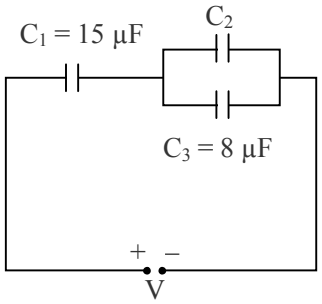
$T_1 = 2g$

$\frac{V_1}{V_2} = \frac{\lambda_1}{\lambda_2}$

$$\lambda_2 = \frac{V_2}{V_1}\lambda_1 = \sqrt{\frac{T_2}{T_1}} \times \lambda_1$$

$$= \sqrt{\frac{8g}{2g}}\lambda_1 = 2 \times 6 = 12 \text{ cm}$$

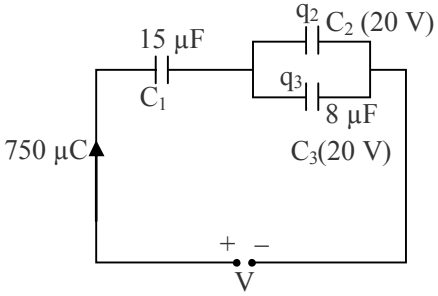
Q.9 In the circuit shown in the figure, the total charge in 750 μC and the voltage across capacitor C_2 is 20 V. Then the charge on capacitor C_2 is



(1) 450 μC (2) 590 μC (3) 160 μC (4) 650 μC

Ans. **[2]**

Sol.



$q_3 = 20 \times 8 = 160$ μC

$\therefore\ q_2 = 750 - 160 = 590$ μC

Q.10 A 750 Hz, 20 V (rms) source is connected to a resistance of 100 Ω, an inductance of 0.1803 H and a capacitance of 10 μF all in series. The time in which the resistance (heat capacity 2 J/ºC) will get heated by 10ºC. (assume no loss of heat to the surroundings) is close to-

(1) 365 s (2) 418 s (3) 245 s (4) 348 s

Ans. **[4]**

Sol. f = 750 Hz, V_{rms} = 20 V,

R = 100 Ω, L = 0.1803 H,

C = 10μ F, S = 2 J/°C

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$$

$$= \sqrt{R^2 + \left(2\pi fL - \frac{1}{2\pi fC}\right)^2}$$

Putting values

$|Z| = 834\Omega$

In AC power $P = V_{rms}\, i_{rms} \cos\phi$

$\cos\phi = \dfrac{R}{|Z|}$ $\quad i_{rms} = \dfrac{V_{rms}}{|Z|}$

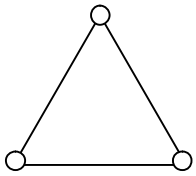
$= \dfrac{V_{rms}^2\, R}{(|Z|)^2}$

$= \left(\dfrac{20}{834}\right)^2 \times 100 = 0.0575$ J/s

$H = Pt = S\Delta\theta$

$t = \dfrac{2(10)}{0.0575} = 348$ sec

Q.11 Consider a gas of triatomic molecules. The molecules are assumed to be traingular and made of massless rigid rods whose vertices are occupied by atoms. The internal energy of a mole of the gas at temperature T is-



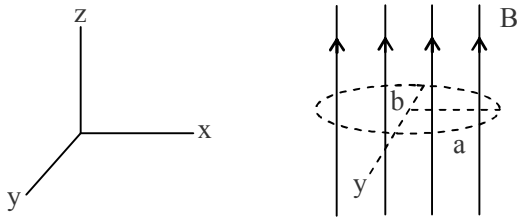
(1) 3RT (2) $\dfrac{5}{2}$RT (3) $\dfrac{9}{2}$RT (4) $\dfrac{3}{2}$RT

Ans. **[1]**

Sol. DOF = 3 + 3 = 6

$U = \dfrac{f}{2}nRT = 3RT$

Q.12 An elliptical loop having resistance R, of semi major axis a, and semi minor axis b is placed in a magnetic field as shown in the figure. If the loop is rotated about the x-axis with angular frequency ω, the average power loss in the loop due to Joule heating is



(1) zero (2) $\dfrac{\pi^2 a^2 b^2 B^2 \omega^2}{R}$ (3) $\dfrac{\pi^2 a^2 b^2 B^2 \omega^2}{2R}$ (4) $\dfrac{\pi abB\omega}{R}$

Ans. **[3]**

Sol. $\in = NAB\omega\cos\omega t$ $\qquad N = 1$

$P_{avg} = <\dfrac{\in^2}{R}> = <\dfrac{(AB\omega\cos\omega t)^2}{R}>$

$= \dfrac{A^2B^2\omega^2}{R}\ \dfrac{1}{2} = \dfrac{\pi^2 a^2 b^2 B^2 \omega^2}{2R}$

Q.13 Pressure inside two soap bubbles are 1.01 and 1.02 atmosphere, respectively. the ratio of their volumes is

(1) 4 : 1 (2) 2 : 1 (3) 8 : 1 (4) 0.8 : 1

Ans. **[3]**

Sol. $\Delta P_1 = 0.01 = 4T/R_1$ .....(1)

$\Delta P_2 = 0.02 = 4T/R_2$ ....(2)

Equation (1) ÷ (2)

$$\frac{1}{2} = \frac{R_2}{R_1}$$

$R_1 = 2R_2$

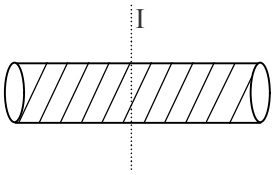
$$\frac{V_1}{V_2} = \frac{R_1^3}{R_2^3} = \frac{8R_2^3}{R_2^3} = 8$$

Q.14 Moment of inertia of a cylinder of mass M, length L and radius R about an axis passing through its centre and perpendicular to the axis of the cylinder is $I = M\left(\frac{R^2}{4} + \frac{L^2}{12}\right)$. If such a cylinder is to be made for a given mass of a material, the ratio L/R for it to have minimum possible I is-

(1) $\frac{2}{3}$ (2) $\frac{3}{2}$ (3) $\sqrt{\frac{2}{3}}$ (4) $\sqrt{\frac{3}{2}}$

Ans. **[4]**

Sol.



$$I = M\left(\frac{R^2}{4} + \frac{L^2}{12}\right) \quad ....(1)$$

as mass is constant $\Rightarrow m = \rho V$ = constant

V = constant

$\pi^2 R\ell$ = constant $\Rightarrow R^2 L$ = constant

$$2RL + R^2\frac{dL}{dR} = 0 \quad ...(2)$$

From equation (1)

$$\frac{dI}{dR} = M\left(\frac{2R}{4} + \frac{2L}{12} \times \frac{dL}{dr}\right) = 0$$

$$\frac{R}{2} + \frac{L}{6}\frac{dL}{dR} = 0$$

Substituting value of $\frac{dL}{dR}$ from equation (2)

$$\frac{R}{2}+\frac{L}{6}\left(\frac{-2L}{R}\right)=0$$

$$\frac{R}{2}+\frac{L^2}{3R}\Rightarrow\frac{L}{R}=\sqrt{\frac{3}{2}}$$

Q.15 Using screw gauge of pitch 0.1 cm and 50 divisions on its circular scale, the thickness of an object is measured. It should correctly be recorded as-

(1) 2.124 cm (2) 2.121 cm (3) 2.125 cm (4) 2.123 cm

Ans. **[1]**

Sol. $LC=\frac{\text{pitch}}{CSD}=\frac{0.1\text{cm}}{50}=0.002$ cm

So any measurement will be integral

Multiple of LC.

So ans. will be 2.124 cm

Q.16 In a radioactive material, fraction of active material remaining after time t is 9/16. The fraction that was remaining after t/2 is-

(1) $\frac{3}{5}$ (2) $\frac{7}{8}$ (3) $\frac{3}{4}$ (4) $\frac{4}{5}$

Ans. **[3]**

Sol. First order decay

$N(t)=N_0e^{-\lambda t}$

Given $N(t)/N_0=9/16=e^{-\lambda t}$

Now, $N(t/2)=No\ e^{-\lambda\ t/2}$

$$\frac{N(t/2)}{N_0}=\sqrt{e^{-\lambda t}}=\sqrt{9/16}$$

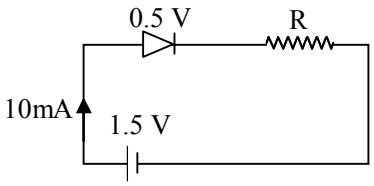
$N(t/2)=3/4\ N_0$

Q.17 When a diode is forward biased, it has a voltage drop of 0.5 V. The safe limit of current through the diode is 10 mA. If a battery of emf 1.5 V is used in the circuit, the value of minimum resistance to be connected in series with the diode so that the current does not exceed the safe limit is-

(1) 300 Ω (2) 50 Ω (3) 100 Ω (4) 200 Ω

Ans. **[3]**

Sol.



$1.5-0.5-R\times10\times10^{-3}=0$

$\therefore\ R=100\ \Omega$

Q.18 A charged particle carrying charge 1 μC is moving with velocity $(2\hat{i}+3\hat{j}+4\hat{k})$ m/s. If an external magnetic field of $(5\hat{i}+3\hat{j}-6\hat{k}) \times 10^{-3}$ T, exists in the region where the particle is moving then the force on the particle is $\vec{F} \times 10^{-9}$ N. The vector $\vec{F}$ is-

(1) $-3.0\hat{i}+3.2\hat{j}-0.9\hat{k}$ (2) $-300\hat{i}+320\hat{j}-90\hat{k}$

(3) $-30\hat{i}+32\hat{j}-9\hat{k}$ (4) $-0.30\hat{i}+0.32\hat{j}-0.09\hat{k}$

Ans. **[3]**

Sol. $\vec{F} = 9\ (\vec{V}\times\vec{B})$ (Force on charge particle moving in magnetic field)

$\vec{V}\times\vec{B} = (2\hat{i}+3\hat{j}+4\hat{k}) \times (5\hat{i}+3\hat{j}-6\hat{k}) \times 10^{-3}$

$$= \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 5 & 3 & -6 \end{pmatrix} \times 10^{-3}$$

$= \left[\hat{i}[-18-12]-\hat{j}[-12-20]+\hat{k}[6-15]\right]\times 10^{-3}$

$= [i[-30]+\hat{j}[32]+\hat{k}[-9]\times 10^{-3}$

Force $= 10^{-6}\ [-30\hat{i}+32\hat{j}-9\hat{k}] \times 10^{-3}$

$= 10^{-9}[-30\hat{i}+32\hat{j}-9\hat{k}]$

Q.19 The magnetic field of a plane electromagnetic wave is

$\vec{B} = 3\times10^{-8}\sin[200\pi(y+ct)]\hat{i}$ T

where c = 3 × 10^8 m/s is the speed of light.

The corresponding electric field is-

(1) $\vec{E} = -10^{-6}\sin[200\pi(y+ct)]\hat{k}$ V / m (2) $\vec{E} = 3\times10^{-8}\sin[200\pi(y+ct)]\hat{k}$ V / m

(3) $\vec{E} = -9\sin[200\pi(y+ct)]\hat{k}$ V / m (4) $\vec{E} = 9\sin[200\pi(y+ct)]\hat{k}$ V / m

Ans. **[3]**

Sol. $\vec{B} = 3 \times 10^{-8} \sin [200\ \pi(y + ct)]\ \hat{i}$ T

$E_0 = CB_0 \Rightarrow E_0 = 3 \times 10^8 \times 3 \times 10^{-8}$

$= 9$ V/m

and direction of wave propagation is given as

$(\vec{E}\times\vec{B}) \parallel \vec{C}$

$\hat{B} = \hat{i}$ & $\hat{C} = -\hat{j}$

so $\hat{E} = -\hat{k}$

$\therefore\ \vec{E} = E_0 \sin [200\pi\ (y + ct)]\ (-\hat{k})$ V/m

Q.20 In a Young's double slit experiment, light of 500 nm is used to produce an interference pattern. When the distance between the slits is 0.05 mm, the angular width (in degree) of the fringes formed on the distance screen is close to-

(1) 1.7º (2) 0.07º (3) 0.57º (4) 0.17º

Ans. **[3]**

Sol. $\Delta\theta_0 = \left(\frac{\lambda}{d}\times\frac{180}{\pi}\right)^0$

$= 0.57°$

Q.21 When a long glass capillary tube of radius 0.015 cm is dipped in a liquid, the liquid rises to a height of 15 cm within it. If the contact angle between the liquid and glass to close to 0º, the surface tension of the liquid, in milliNewton m^{-1}, is [$\rho_{(liquid)} = 900\ kgm^{-3}$, $g = 10\ ms^{-2}$] (Give answer in closest integer) _____ .

Ans. **[101]**

Sol. Capillary rise

$h = \frac{2S\cos\theta}{\rho gr}$

$S = \frac{\rho grh}{2\cos\theta}$

$= \frac{(900)(10)(15\times10^{-5})(15\times10^{-2})}{2}$

$S = 1012.5 \times 10^{-4}$

$S = 101.25 \times 10^{-3} = 101.25$ mN/m

In question closest integer is asked so closest integer = 101.00 Ans.

Q.22 A person of 80 kg mass is standing on the rim of a circular platform of mass 200 kg rotating about its axis as 5 revolutions per minute (rpm). The person now starts moving towards the centre of the platform. What will be the rotational speed (in rpm) of the platform when the person reaches its centre _____ .

Ans. **[9]**

Sol. $L_i = L_f$

$\left(80R^2 + \frac{200R^2}{2}\right)\omega = \left(0 + \frac{200R^2}{2}\right)\omega_1$

$180\omega_0 = 100\omega_1$

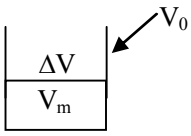
$\omega_1 = 1.8\omega_0 = 1.8 \times 5$

$= 9$ rpm

Q.23 A bakelite beaker has volume capacity of 500 cc at 30ºC. When it is partially filled with V_m volume (at 30ºC) of mercury, it is found that the unfilled volume of the beaker remains constant as temperature is varied. If $\gamma_{(beaker)} = 6 \times 10^{-6}\ ºC^{-1}$ and $\gamma_{(mercury)} = 1.5 \times 10^{-4}\ ºC^{-1}$, where γ is the coefficient of volume expansion, then V_m (in cc) is close to _____ .

Ans. **[20]**

Sol.



$\Delta V = (V_0 - V_m)$

After increasing temperature

$\Delta V' = (V'_0 - V'_m)$

$\Delta V' = \Delta V$

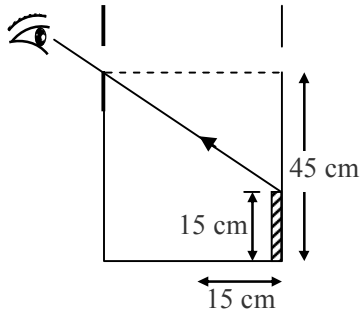
$V_0 - V_m = V_0(1+ \gamma_b \Delta T) - V_m (1 + \gamma_M \Delta T)$

$V_0\gamma_b = V_m\gamma_m$

$$V_m = \frac{V_0\gamma_b}{\gamma_m} = \frac{(500)(6\times10^{-6})}{(1.5\times10^{-4})}$$

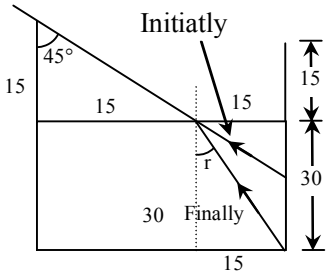
$= 20$ CC

Q.24 An observer can see through a small hole on the side of a jar (radius 15 cm) at a point at height of 15 cm from the bottom (see figure). The hole is at a height of 45 cm. When the jar is filled with a liquid up to a height of 30 cm the same observer can see the edge at the bottom of the jar. If the refractive index of the liquid is N/100, where N is an integer, the value of N is _____.



Ans. **[158]**

Sol.



$$\tan r = \frac{15}{30} = \frac{1}{2}$$

$$\sin r = \frac{1}{\sqrt{5}}$$

$1 \sin 45° = \mu \sin r$

$$\frac{1}{\sqrt{2}} = \mu\left(\frac{1}{\sqrt{5}}\right)$$

$$\mu = \sqrt{\frac{5}{2}} = 1.581$$

$$\frac{N}{100} = \mu$$

N = 100 m

N = 158.11

So integer value of N = 15800

Q.25 A cricket ball of mass 0.15 kg is thrown vertically up by a bowling machine so that it rises to a maximum height of 20 m after leaving the machine. If the part pushing the ball applies a constant force F on the ball and moves horizontally a distance of 0.2 m while launching the ball, the value of F (in N) is ($g = 10\ m/s^2$) _____ .

Ans. **[150]**

Sol. $W_F = \frac{1}{2}mv^2 = mgh$

F(S) = mgh

F(0.2) = (0.15) (10) (20)

F = 150 N

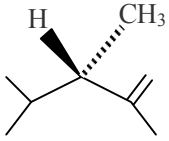
JEE Main Online Exam 2020

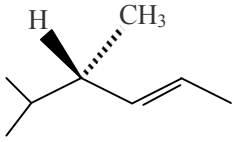
Question with Solutions

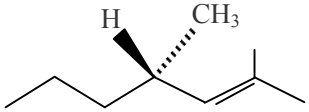
3rd September 2020 | Shift-I

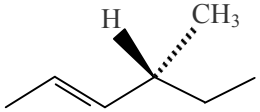
CHEMISTRY

Q.1 Which of the following compounds produces an optically inactive compound on hydrogenation ?

(1) 

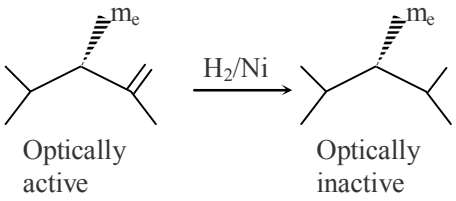
(2) 

(3) 

(4) 

Ans. **[1]**

Sol.



Q.2 The electronic spectrum of $[Ti(H_2O)_6]^{3+}$ shows a single broad peak with a maximum at 20,300 cm^{-1}. The crystal field stabilization energy (CFSE) of the complex ion, in kJ mol^{-1}, is-

(1) 97 (2) 145.5 (3) 83.7 (4) 242.5

Ans. **[4]**

Sol. CFSE = 0.4 Δ_0

$= 0.4 \times \frac{20300}{83.7}$

= 97 kJ/mol.

Q.3 An acidic buffer is obtained on mixing :

(1) 100 mL of 0.1 M CH_3COOH and 100 mL of 0.1 M NaOH

(2) 100 mL of 0.1 M HCl and 200 mL of 0.1 M NaCl

(3) 100 mL of of 0.1 M HCl and 200 mL of 0.1 M CH_3COONa

(4) 100 mL of 0.1 M CH_3COOH and 200 mL of 0.1 M NaOH

Ans. **[3]**

Sol.

$$HCl + CH_3COONa \rightarrow CH_3COOH + NaCl$$

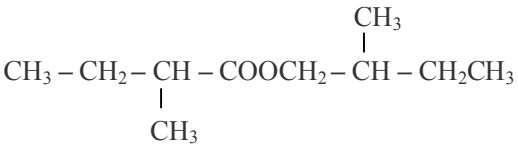
10 mili mol	20 mili mol	—	—
—	10 mili mol	10 mili mol	10 mili mol

So finally we get mixture of

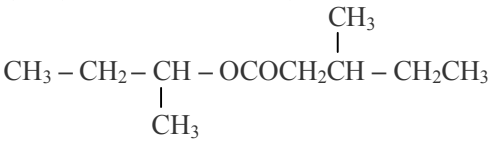
$CH_3COOH + CH_3COONa$ that will work like acidic buffer solution.

Q.4 An organic compound [A], molecular formula $C_{10}H_{20}O_2$ was hydrolyzed with dilute sulphuric acid to give a carboxylic acid [B] and an alcohol [C]. Oxidation of [C] with $CrO_3 - H_2SO_4$ produced [B]. Which of the following structure are not possible for [A] ?

(1) $CH_3CH_2CH_2COOCH_2CH_2CH_2CH_3$

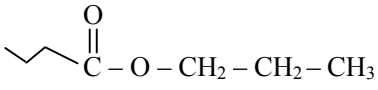
(2) 

(3) $(CH_3)_3 - C - COOCH_2C(CH_3)_3$

(4) 

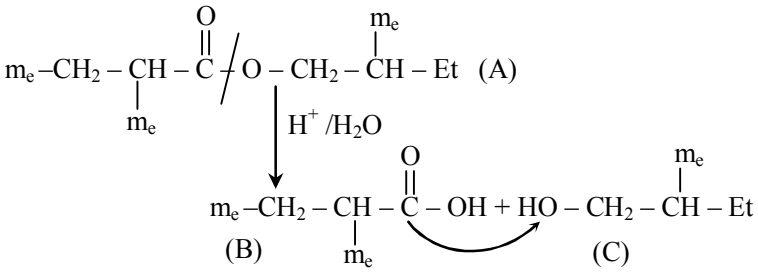
Ans. **[4]**

Sol.

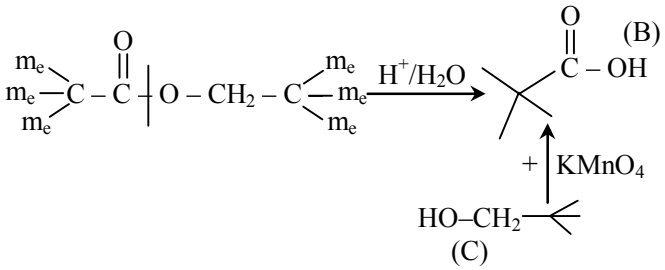
(1) 

Total 8 'C' $\rightarrow$ so molecular formula not matched.

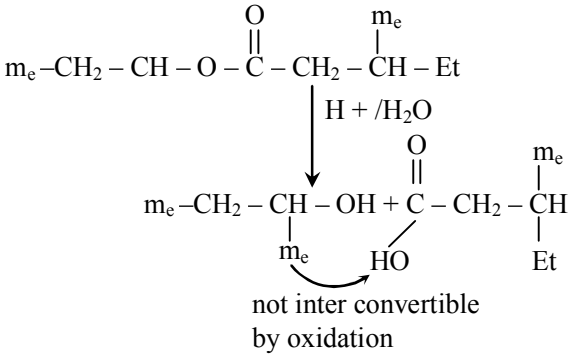
(2)



(3)



(4)



Q.5 Aqua regia is used for dissolving noble metals (Au, Pt, etc.). The gas evolved in this process is-

(1) N_2O_5 (2) N_2O_3 (3) N_2 (4) NO

Ans. **[4]**

Sol. $Au + HNO_3 + 4HCl \rightarrow HAuCl_4 + NO + 2H_2O$

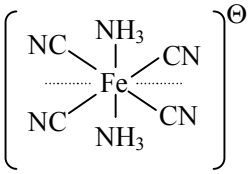
Q.6 The complex that can show optical activity is-

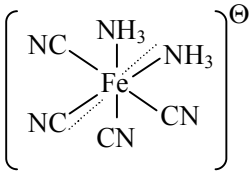
(1) *trans*-$[Fe(NH_3)_2(CN)_4]^-$ (2) *cis*-$[CrCl_2(ox)_2]^{3-}$ (ox = oxalate)

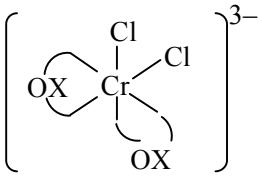
(3) *cis*-$[Fe(NH_3)_2(CN)_4]^-$ (4) *trans*-$[Cr(Cl_2(ox)_2]^{3-}$

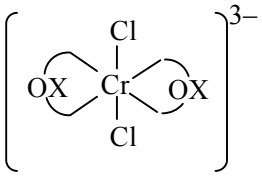
Ans. **[2]**

Sol.

(1)  optically inactive

(2)  optically inactive

(3)  optically active

(4)  optically inactive

Q.7 Thermal power plants can lead to-

(1) Acid rain (2) Eutrophication

(3) Blue baby syndrome (4) Ozone layer depletion

Ans. **[1]**

Sol. Thermal power plants lead to acid rain.

Q.8 Henry's constant (in kbar) for four gases α, β, γ and δ in water at 298 K is given below :

	α	β	γ	δ
K_H	50	2	2×10^{-5}	0.5

(density of water = 10^3 kg m^{-3} at 298 K)

This table implies that -

(1) α has the highest solubility in water at a given pressure

(2) solubility of γ at 308 K is lower than at 298 K

(3) The pressure of a 55.5 molal solution of γ is 1 bar

(4) The pressure of a 55.5 molal solution of δ is 250 bar

Ans. **[4]**

Sol. $P_\gamma = K_H X_Y$

$$P_\gamma = 2 \times 10^{-15} \times \frac{55.5}{55.5 + \frac{1000}{18}} = 2 \times 10^{-5} \text{ K bar}$$

$= 2 \times 10^{-2}$ bar

(2) $P_\delta = K_H X_\delta$

$$P_\delta = 0.5 \times \frac{55.5}{55.5 + \frac{1000}{18}} = 249 \text{ K bar} = 249 \text{ bar}$$

(3) On increasing temperature solubility of gases decreases

(4) $K_H \downarrow$ solubility $\uparrow$ and lowest K_H is for γ.

Q.9 Of the species, NO, NO^+, NO^{2+} and NO^-, the one with minimum bond strength is-

(1) NO^{2+} (2) NO^+ (3) NO (4) NO^{-0}

Ans. **[4]**

Sol. Bond order of $NO^{2+} = 2.5$

Bond order of $NO^+ = 3$

Bond order of NO = 2.5

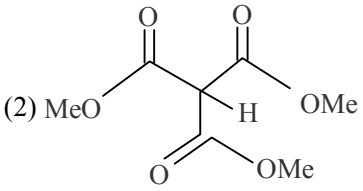
Bond order of $NO^- = 2$

Bond order α bond strength

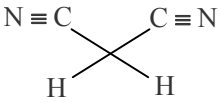
Q.10 Which one of the following compounds possesses the most acidic hydrogen ?

(1) $H_3C - C \equiv C - H$

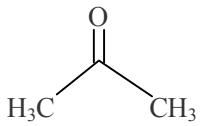
(2)



(3)

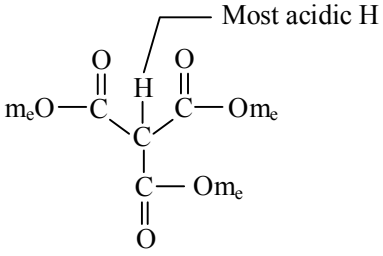


(4)



Ans. **[2]**

Sol.



Due to presence of 3 (–R) groups

Q.11 The atomic number of the element unnilennium is-

(1) 108 (2) 109 (3) 102 (4) 119

Ans. **[2]**

Sol. 1 0 9

un nil enn

Hence correct name → unnilennium

Q.12 Glycerol is separated in soap industries by-

(1) Steam Distillation (2) Distillation under reduced pressure

(3) Fractional distillation (4) Differential extraction

Ans. **[2]**

Sol. Glycerol is separated by reduced pressure distillation in soap industries.

Q.13 If the boiling point of H_2O is 373 K, the boiling point of H_2S will be-

(1) greater than 300 K but less than 373 K

(2) less than 300 K

(3) more than 373 K

(4) equal to 373 K

Ans. **[2]**

Sol. Boiling point of H_2S < Boiling point of H_2O

(213 K) (373 K)

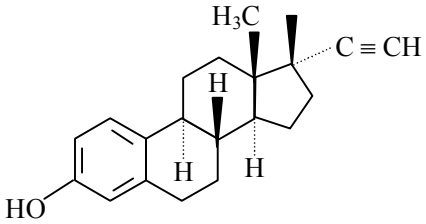
Q.14 The antifertility drug "Novestrol" can react with-

(1) Br_2/water ; $ZnCl_2$/HCl ; $FeCl_3$ (2) Alcoholic HCN ; NaOCl ; $ZnCl_2$/HCl

(3) $ZnCl_2$/HCl ; $FeCl_3$; Alcoholic HCN (4) Br_2/water ; $ZnCl_2$/HCl ; NaOCl

Ans. **[1]**

Sol.



Ethynylestradiol (novestrol)

gives

(1) $Br_2 + H_2O$ test

(2) Lucas test with $ZnCl_2$ + HCl

(3) $FeCl_3$ test of phenolic group

Q.15 Tyndall effect is observed when-

(1) The diameter of dispersed particles is similar to the wavelength of light used

(2) the diameter of dispersed particles is much smaller than the wavelength of light used

(3) The refractive index of dispersed phase is greater than that of the dispersion medium

(4) The diameter of dispersed particles is much larger than the wavelength of light used

Ans. **[1]**

Sol. The diameter of disperseed particles is similar to wavelength of light used.

Q.16 The mechanism of S_N^1 reaction is given as :

$$R - X \longrightarrow \underset{\text{Ion pair}}{R^{\oplus}X^{\ominus}} \longrightarrow \underset{\text{Solvent separated ion pair}}{R^{\oplus} \,||\, X^{\ominus}} \xrightarrow{Y^{\ominus}} R - Y + X^{\ominus}$$

A student writes general characteristics based on the given mechanism as-

(a) The reaction is favoured by weak nucleiphiles

(b) $R^{\oplus}$ would be easily formed if the substituents are bulky

(c) The reaction is accompained by racemization

(d) The reaction is favoured by non-polar solvents

Which observations are correct ?

(1) (a) and (d) (2) (a) and (c) (3) (a) and (b) (4) (a), (b) and (c)

Ans. **[2]**

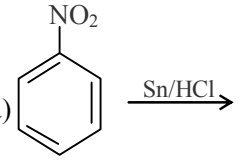
Sol. S_N^1 favours

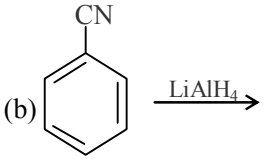
(a) The reaction is favoured by weak nucleophiles

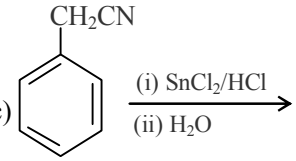
(b) $R^{\oplus}$ would be easily formed if the substituents are bulky

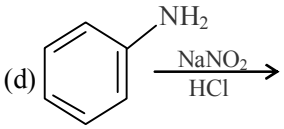
(c) The reaction is accoumpained by recemization

Q.17 The Kjeldahl method of Nitrogen estimation fails for which of the following reaction products ?

(a) 

(b) 

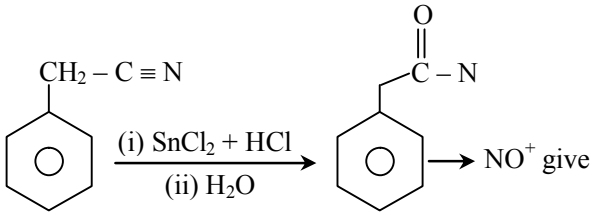
(c) 

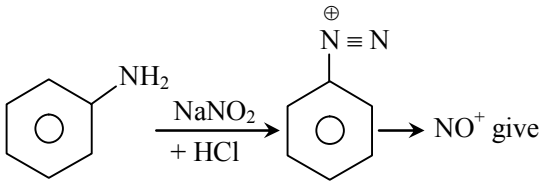
(d) 

(1) (a), (c) and (d) (2) (b) and (c) (3) (a) and (d) (4) (c) and (d)

Ans. **[4]**

Sol. Kjeldahl method is used for N estimation But not given by 'Diazo' compounds.





Q.18 Let C_{NaCl} and C_{BaSO_4} be the conductances (in S) measured for saturated aqueous solutions of NaCl and $BaSO_4$, respectively, at a temperature T.
Which of the following is false ?
(1) $C_{NaCl} >> C_{BaSO_4}$ at a given T
(2) $C_{NaCl}(T_2) > C_{NaCl}(T_1)$ for $T_2 > T_1$
(3) Ionic mobilities of ions from both salts increase with T
(4) $C_{BaSO_4}(T_2) > C_{BaSO_4}(T_1)$ for $T_2 > T_1$

Ans. **[2]**

Sol. Dissolution of $BaSO_4$ is and endothermic reaction 50 on increasing temperature number of ions of $BaSO_4$ decrease so it's conduction also decrease.

Q.19 It is true that :
(1) A second order reaction is always a multistep reaction
(2) A zero order reaction is a single step reaction
(3) A first order reaction is always a single step reaction
(4) A zero order reaction is a multistep reaction

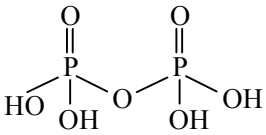
Ans. **[4]**

Sol. Zero order reaction is multiple step reaction.

Q.20 In a molecule of pyrophosphoric acid, the number of P – OH, P = O and P – O – P bonds/moiety (ies) respectively are :
(1) 4, 2 and 0
(2) 2, 4 and 1
(3) 4, 2 and 1
(4) 3, 3 and 3

Ans. **[3]**

Sol. Pyrophosphoric acid.



P – OH linkages = 4
P = O linkages = 2
P – O – P linkages = 1

Q.21 The mole fraction of glucose ($C_6H_{12}O_6$) in an aqueous binary solution is 0.1. The mass percentage of water in it, to the nearest integer, is _____ .

Ans. **[47]**

Sol. $X_{C_6H_{12}O_6} = 0.1$
Let total mole is 1 mol then mole of glucose will be 0.1 and mole of water will be 0.9

$$\text{so mass \% of water} = \frac{0.9\times18}{0.1\times180+0.9\times18}\times100$$
$$= 47.36$$

Q.22 The volume strength of 8.9 M H_2O_2 solution calculated at 273 K and 1 atm is _____. (R = 0.0821 L atm K^{-1} mol^{-1}) rounded off to the nearest integer)

Ans. **[100]**

Sol. Volume strength of H_2O_2 at 1 atm

273 kelvin = M × 11.2 = 8.9 × 11.2 = 99.68

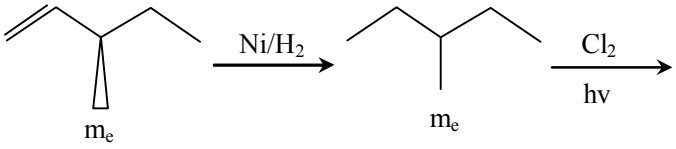
Ans. : 100

Q.23 The total number of monohalogenated organic products in the following (including stereoisomers) reaction is _____ .

A (Simplest optically active alkene) $\xrightarrow[\text{(ii) } X_2/\Delta]{\text{(i) } H_2/Ni/\Delta}$

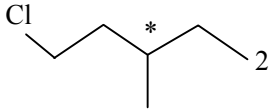
Ans. **[8]**

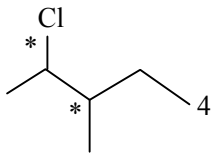
Sol.

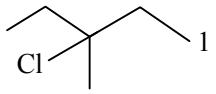


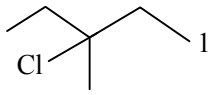
Simplest

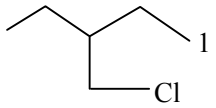
O. A. Alkene





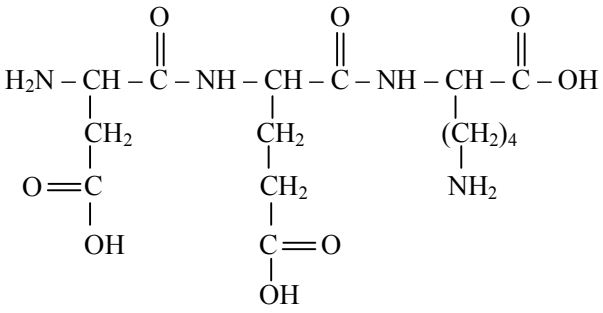






Alter

str. of Tri peptide



Q.24 The photoelectric current from Na (work function, $w_0 = 2.3$ eV) is stopped by the output voltage of the cell Pt(s) | H_2(g, 1 bar) | HCl (aq., pH = 1) | AgCl(s) | Ag(s).
The pH of aq. HCl required to stop the photoelectric current from K($w_0 = 2.25$ eV), all other conditions remaining the same, is _____ $\times 10^{-2}$ (to the nearest integer).

Given $2.303\ \frac{RT}{F} = 0.06$ V ; $E^0_{AgCl|Ag|Cl^-} = 0.22$ V

Ans. **[1.42]** NTA official Ans. 58

Sol. $\frac{1}{2}H_2 \rightarrow H^+ + e^\Theta$

$e^\Theta + AgCl_{(s)} \rightarrow Ag_{(s)} + Cl^\Theta$

$\frac{1}{2}H_2 + AgCl_{(s)} \rightarrow H^+_{(aq)} + Ag_{(s)} + C^\Theta_{(aq)}$

$$E = \varepsilon^0 - \frac{.06}{1}\log\frac{[H^+]\,[Cl^\Theta]}{P_{H_2}^{\frac{1}{2}}}$$

$$E = 0.22 - .06\log\frac{(10^{-1})(10^{-1})}{1^{\frac{1}{2}}}$$

$E = 0.22 + .12 = .34$ volt

$\Rightarrow$ total energy of photon will be (for Na) = 2.3 + 0.34 = 2.64 eV

$\Rightarrow$ stopping potential required for K

= 2.64 – 2.25 = 0.39 volt

$$E = \varepsilon^0 - \frac{.06}{1}\log\frac{[H^+]\,[Cl^-]}{P_{H_2}^{\frac{1}{2}}}$$

as $[H^+] = [Cl^\Theta]$ so

$$0.39 = 0.22 - .06\log\frac{[H^+]^2}{1^{\frac{1}{2}}}$$

$0.17 = +.12$ pH

pH = 1.4155 $\Rightarrow$ 1.42

Q.25 An element with molar mass 2.7×10^{-2} kg mol^{-1} forms cubic unit cell with edge length 405 pm. If its density is 2.7×10^3 kg m^{-3}, the radius of the element is approximately _____ $\times 10^{-12}$ m (to the nearest integer).

Ans. **[143]**

Sol. $$d = \frac{z\left(\frac{M}{N_A}\right)}{a^3}$$

$$2.7 \times 10^3 = z\,\frac{\left(\frac{2.7\times10^{-2}}{6\times10^{23}}\right)}{(405\times10^{-12})^3}$$

$$2.7 \times 10^3 = z \frac{(2.7\times10^{-2})}{6\times10^{23}(4.05\times10^{-10})^3}$$

$$2.7 \times 10^3 = z \frac{(2.7\times10^{-2})}{6\times10^{23}\times66.43\times10^{-30}}$$

$3.98 = z$

$z \approx 4$ structure is fcc

$$\frac{a}{\sqrt{2}} = 2r$$

$$r = \frac{a}{2\sqrt{2}} = \frac{\sqrt{2}a}{4} = \frac{1.414\times405\times10^{-12}}{4}$$

$r = 143.16 \times 10^{-12}$

JEE Main Online Exam 2020

Question with Solutions

3rd September 2020 | Shift-I

MATHEMATICS

Q.1 The proposition $p \rightarrow \sim(p \wedge \sim q)$ is equivalent to

(1) $(\sim p) \vee q$ (2) $(\sim p) \vee (\sim q)$ (3) q (4) $(\sim p) \wedge q$

Ans. **[1]**

Sol. $p \rightarrow \sim (p \wedge \sim q)$

$= \sim p \vee \sim (p \wedge \sim q)$

$= \sim p \vee \sim p \vee q$

$= \sim (p \wedge q) \vee q$

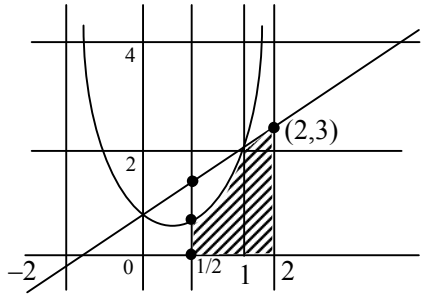
$= \sim p \vee q$

Q.2 The area (in sq. units) of the region $\{(x, y) : 0 \le y \le x^2 + 1, 0 \le y \le x + 1, \frac{1}{2} \le x \le 2\}$ is

(1) $\frac{23}{6}$ (2) $\frac{79}{24}$ (3) $\frac{79}{16}$ (4) $\frac{23}{16}$

Ans. **[2]**

Sol. $0 \le y \le x^2 + 1, 0 \le y \le x + 1, \frac{1}{2} \le x \le 2$



Required area $= \int_{1/2}^{1} (x^2 + 1)dx + \frac{1}{2}(2 + 3) \times 1$

$= \frac{19}{24} + \frac{5}{2} = \frac{79}{24}$

Q.3 If the number of integral terms in the expansion of $(3^{1/2} + 5^{1/8})^n$ is exactly 33, then the least value of n is-

(1) 128 (2) 264 (3) 256 (4) 248

Ans. **[3]**

Sol. $T_{r+1} = {}^{n}C_{r}\,(3)^{\frac{n-r}{2}}\,(5)^{\frac{r}{8}}$ $(n \geq r)$

Clearly r should be a multiple of 8.

$\therefore$ there are exactly 33 integral terms

possible values of r can be

0, 8, 16, ..., 32×8

$\therefore$ least value of n = 256.

Q.4 $2\pi - \left(\sin^{-1}\frac{4}{5} + \sin^{-1}\frac{5}{13} + \sin^{-1}\frac{16}{65}\right)$ is equal to-

(1) $\frac{3\pi}{2}$ (2) $\frac{7\pi}{4}$ (3) $\frac{\pi}{2}$ (4) $\frac{5\pi}{4}$

Ans. **[1]**

Sol. $2\pi - \left(\sin^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{5}{13}\right) + \sin^{-1}\left(\frac{16}{65}\right)\right)$

$= 2\pi - \left(\tan^{-1}\left(\frac{4}{3}\right) + \tan^{-1}\left(\frac{5}{12}\right) + \tan^{-1}\left(\frac{16}{63}\right)\right)$

$= 2\pi - \left(\tan^{-1}\left(\frac{63}{16}\right) + \tan^{-1}\left(\frac{16}{63}\right)\right)$

$= 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$

Q.5 The lines

$\vec{r} = (\hat{i} - \hat{j}) + l(2\hat{i} + \hat{k})$ and

$\vec{r} = (2\hat{i} - \hat{j}) + m(\hat{i} + \hat{j} - \hat{k})$

(1) intersect for all values of l and m

(2) do not intersect for any values of l and m

(3) intersect when $l = 2$ and $m = \frac{1}{2}$

(4) intersect when $l = 1$ and $m = 2$

Ans. **[2]**

Sol. $\vec{r} = \hat{i}(1 + 2\ell) + \hat{j}(-1) + \hat{k}(\ell)$

$\vec{r} = \hat{i}(2 + m) + \hat{j}(m - 1) + \hat{k}(-m)$

For intersection

$1 + 2\ell = 2 + m$ ....(i)

$-1 = m - 1$ ....(ii)

$\ell = -m$ .....(iii)

from (ii) m = 0

from (iii) $\ell = 0$

These values of m and ℓ do not satisfy equation (1).

Hence the two lines do not intersect for any values of ℓ and m

Q.6 $\int_{-\pi}^{\pi} |\pi - |x|| \, dx$ is equal to-

(1) π^2 (2) $\frac{\pi^2}{2}$ (3) $2\pi^2$ (4) $\sqrt{2}\,\pi^2$

Ans. **[1]**

Sol. $\int_{-\pi}^{\pi} |\pi - |x|| \, dx = 2\int_{0}^{\pi} |\pi - x| \, dx$

$= 2\int_{0}^{\pi} (\pi - x)dx$

$= 2\left[\pi x - \frac{x^2}{2}\right]_0^{\pi} = \pi^2$

Q.7 Consider the two sets :

$A = \{m \in R$: both the roots of $x^2 - (m+1)x + m + 4 = 0$ are real$\}$ and $B = \{-3, 5\}$.

Which of the following is not true ?

(1) $A \cap B = \{-3\}$ (2) $A \cup B = R$

(3) $A - B = (-\infty, -3) \cup (5, \infty)$ (4) $B - A = (-3, 5)$

Ans. **[3]**

Sol. $A : B \geq 0$

$\Rightarrow (m+1)^2 - 4(m+4) \geq 0$

$\Rightarrow m^2 + 2m + 1 - 4m - 16 \geq 0$

$\Rightarrow m^2 - 2m - 15 \geq 0$

$\Rightarrow (m-5)(m+3) \geq 0$

$\Rightarrow m \in (-\infty, -3] \cup [5, \infty)$

$\therefore A = (-\infty, -3] \cup [5, \infty)$

$B = [-3, 5]$

$A - B = (-\infty, -3) \cup [5, \infty)$

$A \cap B = \{-3\}$

$B - A = (-3, 5)$

$A \cup B = R$

Q.8 If α and β are the roots of the equation $x^2 + px + 2 = 0$ and $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots of the equation $2x^2 + 2qx + 1 = 0$, then $\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$ is equal to-

(1) $\frac{9}{4}(9 - q^2)$ (2) $\frac{9}{4}(9 + q^2)$ (3) $\frac{9}{4}(9 - p^2)$ (4) $\frac{9}{4}(9 + p^2)$

Ans. **[3]**

Sol. α, β are roots of $x^2 + px + 2 = 0$

$\Rightarrow \alpha^2 + p\alpha + 2 = 0$ & $\beta^2 + p\beta + 2 = 0$

$\Rightarrow \frac{1}{\alpha}, \frac{1}{\beta}$ are roots of $2x^2 + px + 1 = 0$

But $\frac{1}{\alpha}, \frac{1}{\beta}$ are roots of $2x^2 + 2qx + 1 = 0$

$\Rightarrow p = 2q$

Also $\alpha + \beta = -p$ $\alpha\beta = 2$

$$\left(\alpha - \frac{1}{\alpha}\right)\left(\beta - \frac{1}{\beta}\right)\left(\alpha + \frac{1}{\beta}\right)\left(\beta + \frac{1}{\alpha}\right)$$

$$= \left(\frac{\alpha^2 - 1}{\alpha}\right)\left(\frac{\beta^2 - 1}{\beta}\right)\left(\frac{\alpha\beta + 1}{\beta}\right)\left(\frac{\alpha\beta + 1}{\alpha}\right)$$

$$= \frac{(-p\alpha - 3)(-p\beta - 3)(\alpha\beta + 1)^2}{(\alpha\beta)^2}$$

$$= \frac{9}{4}(p\alpha\beta + 3p(\alpha + \beta) + 9)$$

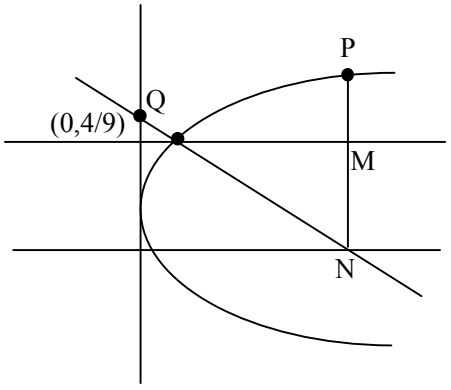
$$= \frac{9}{4}(9 - p^2) = \frac{9}{4}(9 - 4q^2)$$

Q.9 Let P be a point on the parabola, $y^2 = 12x$ and N be the foot of the perpendicular drawn from P on the axis of the parabola A line is now drawn through the mid-point M of PN, parallel to its axis which meets the parabola at Q. If the y-intercept of the line NQ is $\frac{4}{3}$, then-

(1) PN = 4 (2) MQ = $\frac{1}{4}$ (3) PN = 3 (4) MQ = $\frac{1}{3}$

Ans. **[2]**

Sol.



Let P = $(3t^2, 6t)$; N = $(3t^2, 0)$

M = $(3t^2, 3t)$

Equation of MQ : y = 3t

$\therefore$ Q = $\left(\frac{3}{4}t^2, 3t\right)$

Equation of NQ

$$y = \frac{3t}{\left(\frac{3}{4}t^2 - 3t^2\right)}(x - 3t^2)$$

y-intercept of NQ = $4t = \frac{4}{3} \Rightarrow t = \frac{1}{3}$

$\therefore$ MQ = $\frac{9}{4}t^2 = \frac{1}{4}$

PN = 6t = 2

Q.10 If $\Delta = \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix} = Ax^3 + Bx^2 + Cx + D$, then B + C is equal to-

(1) –3 (2) 9 (3) –1 (4) 1

Ans. **[1]**

Sol. $\Delta = \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 2x-3 & 3x-4 & 4x-5 \\ 3x-5 & 5x-8 & 10x-17 \end{vmatrix}$

$= Ax^3 + Bx^2 + Cx + D$

$R_2 \to R_2 - R_1$ $R_3 \to R_3 - R_2$

$\Delta = \begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-1 & x-1 & x-1 \\ x-2 & 2(x-2) & 6(x-2) \end{vmatrix}$

$= (x-1)(x-2)\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ 1 & 1 & 1 \\ 1 & 2 & 6 \end{vmatrix}$

$= -3(x-1)^2(x-2) = -3x^3 + 12x^2 - 15x + 6$

$\therefore$ B + C = 12 – 15 = – 3

Q.11 If the first term of an A.P. is 3 and the sum of its first 25 terms is equal to the sum of its next 15 terms, then the common difference of this A.P. is-

(1) $\frac{1}{7}$ (2) $\frac{1}{4}$ (3) $\frac{1}{5}$ (4) $\frac{1}{6}$

Ans. **[4]**

Sol. Sum of 1st 25 terms = sum of its next 15 terms

$\Rightarrow (T_1 + ... + T_{25}) = (T_{26} + ... + T_{40})$

$\Rightarrow (T_1 + ... + T_{40}) = 2(T_1 + ... + T_{25})$

$\Rightarrow \frac{40}{2}[2 \times 3 + (39\ d)] = 2 \times \frac{25}{2}[2 \times 2 + 24\ d]$

$\Rightarrow d = \frac{1}{6}$

Q.12 A dice is thrown two times and the sum of the scores appearing on the dice is observed to be a multiple of 4. Then the conditional probability that the score 4 has appeared atleast once is-

(1) $\frac{1}{8}$ (2) $\frac{1}{9}$ (3) $\frac{1}{3}$ (4) $\frac{1}{4}$

Ans. **[2]**

Sol. A : Sum obtained is a multiple of 4.

A = {(1, 3), (2, 2), (3, 1), (2, 6), (3, 5), (4, 4), (5, 3), (6, 2), (6, 6)}

B : Score of 4 has appeared at least once.

B = {(1, 4), (2, 4), (3, 4), (4, 4), (5, 4), (6, 4), (4, 1), (4, 2), (4, 3), (4, 5), (4, 6)}

Required probability = $P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)}$

$= \frac{1/36}{9/36} = \frac{1}{9}$

Q.13 The value of $(2 \cdot {}^1P_0 - 3 \cdot {}^2P_1 + 4 \cdot {}^3P_2 - \ldots$ up to 51th term) + (1! – 2! + 3! – .... up to 51th term) is equal to-

(1) 1 + (52) ! (2) 1 – 51(51) ! (3) 1 (4) 1 + (51) !

Ans. **[1]**

Sol. S = (2.${}^1p_0$ – 3. ${}^2P_1$ + 4. ${}^3p_2$ ..... upto 51 terms)

+ (1! + 2! + 3! ..... upto 51 terms)

[$\because {}^np_{n-1} = n!$]

$\therefore$ S = (2 × 1! – 3 × 2! + 4 × 3! ..... 52.51!)

+ (1! – 2! + 3! .....(51) !)

= (2! – 3! + 4! ..... + 52!)

+ (1 ! – 2! + 3! – 4! + ... + (51) !)

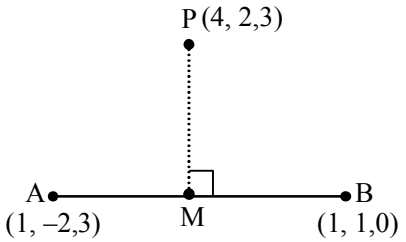
= 1 ! + 52 !.

Q.14 The foot of the perpendicular drawn from the point (4, 2, 3) to the line joining the points (1, –2, 3) and (1, 1, 0) lies on the plane.

(1) x + 2y – z = 1 (2) x – y – 2z = 1 (3) x – 2y + z = 1 (4) 2x + y – z = 1

Ans. **[4]**

Sol.



Equation of AB = $\vec{r} = (\hat{i} + \hat{j}) + \lambda(3\hat{j} - 3\hat{k})$

Let coordinates of M = (1, (1 + 3λ), – 3λ),

$\overrightarrow{PM} = -3\hat{i} + (3\lambda - 1)\hat{j} - 3(\lambda + 1)\hat{k}$

$\overrightarrow{AB} = 3\hat{j} - 3\hat{k}$

$\therefore \overrightarrow{PM} \perp \overrightarrow{AB} \Rightarrow \overrightarrow{PM} \cdot \overrightarrow{AB} = 0$

$\Rightarrow 3(3\lambda - 1) + 9(\lambda + 1) = 0$

$\Rightarrow \lambda = -\frac{1}{3}$

$\therefore M = (1, 0, 1)$

Clearly M lies on $2x + y - z = 1$.

Q.15 Let [t] denote the greatest integer $\le t$. If for some $\lambda \in R - \{0, 1\}$, $\lim\limits_{x\to 0}\left|\frac{1-x+|x|}{\lambda - x + [x]}\right| = L$, then L is equal to-

(1) $\frac{1}{2}$ (2) 0 (3) 2 (4) 1

Ans. **[3]**

Sol. LHL : $\lim\limits_{x\to 0^-}\left|\frac{1-x-x}{\lambda - x - 1}\right| = \left|\frac{1}{\lambda - 1}\right|$

RHL : $\lim\limits_{x\to 0^+}\left|\frac{1-x+x}{\lambda - x + 1}\right| = \left|\frac{1}{\lambda}\right|$

For existence of limit

LHL = RHL

$\Rightarrow \frac{1}{|\lambda - 1|} = \frac{1}{|\lambda|} \Rightarrow \lambda = \frac{1}{2}$

$\therefore L = \frac{1}{|\lambda|} = 2$

Q.16 The function, $f(x) = (3x - 7)x^{2/3}$, $x \in R$, is increasing for all x lying in

(1) $(-\infty, 0) \cup \left(\frac{3}{7}, \infty\right)$ (2) $(-\infty, 0) \cup \left(\frac{14}{15}, \infty\right)$

(3) $\left(-\infty, \frac{14}{15}\right)$ (4) $\left(-\infty, \frac{14}{15}\right) \cup (0, \infty)$

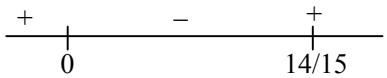
Ans. **[2]**

Sol. $f(x) = (3x - 7)\,x^{2/3}$

$\Rightarrow f(x) = 3x^{5/3} - 7x^{2/3}$

$\Rightarrow f(x) = 5x^{2/3} - \frac{14}{3x^{1/3}}$

$= \frac{15x - 14}{3x^{1/3}} > 0$



$\therefore f'(x) > 0 \ \forall\ x \in (-\infty, 0) \cup \left(\frac{14}{15}, \infty\right)$

Q.17 The solution curve of differential equation, $(1 + e^{-x})(1 + y^2)\frac{dy}{dx} = y^2$, which passes through the point (0, 1) is-

(1) $y^2 + 1 = y\left(\log_e\left(\frac{1+e^x}{2}\right)+2\right)$

(2) $y^2 = 1 + y\log_e\left(\frac{1+e^x}{2}\right)$

(3) $y^2 = 1 + y\log_e\left(\frac{1+e^{-x}}{2}\right)$

(4) $y^2 + 1 = y\left(\log_e\left(\frac{1+e^{-x}}{2}\right)+2\right)$

Ans. **[2]**

Sol. $(1 + e^{-x})(1 + y^2)\frac{dy}{dx} = y^2$

$\Rightarrow (1 + y^{-2})\,dy = \left(\frac{e^x}{1+e^x}\right)dx$

$\Rightarrow \left(y - \frac{1}{y}\right) = \ell n\,(1 + e^x) + c$

$\therefore$ It passes through $(0, 1) \Rightarrow c = -\ell n2$

$\Rightarrow y^2 = 1 + y\,\ell n\left(\frac{1+e^x}{2}\right)$

Q.18 A hyperbola having the transverse axis of length $\sqrt{2}$ has the same foci as that of the ellipse $3x^2 + 4y^2 = 12$, then this hyperbola does not pass through which of the following points ?

(1) $\left(\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}\right)$ (2) $\left(-\sqrt{\frac{3}{2}}, 1\right)$ (3) $\left(\frac{1}{\sqrt{2}}, 0\right)$ (4) $\left(1, -\frac{1}{\sqrt{2}}\right)$

Ans. **[1]**

Sol. Ellipse : $\frac{x^2}{4} + \frac{y^2}{3} = 1$

eccentricity $= \sqrt{1-\frac{3}{4}} = \frac{1}{2}$

$\therefore$ foci $= (\pm 1, 0)$

for hyperbola, given $2a = \sqrt{2} \Rightarrow a = \frac{1}{\sqrt{2}}$

$\therefore$ hyperbola will be

$\frac{x^2}{1/2} - \frac{y^2}{b^2} = 1$

eccentricity $= \sqrt{1+2b^2}$

$\therefore$ foci $= \left(\pm\sqrt{\frac{1+2b^2}{2}}, 0\right)$

$\because$ Ellipse and hyperbola have same foci

$\Rightarrow \sqrt{\frac{1+2b^2}{2}} = 1$

$\Rightarrow b^2 = \frac{1}{2}$

$\therefore$ Equation of hyperbola : $\frac{x^2}{1/2} - \frac{y^2}{1/2} = 1$

$\Rightarrow x^2 - y^2 = \frac{1}{2}$

Clearly $\left(\sqrt{\frac{3}{2}}, \frac{1}{\sqrt{2}}\right)$ does not lie on it.

Q.19 If $y^2 + \log_e(\cos^2 x) = y$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, then-

(1) $|y''(0)| = 2$ (2) $|y'(0)| + |y''(0)| = 3$

(3) $|y'(0)| + |y''(0)| = 1$ (4) $y''(0) = 0$

Ans. **[1]**

Sol. $y^2 + \ln(\cos^2 x) = y$ $\quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

for $x = 0$ $\quad y = 0$ or 1

Differentiating wrt x

$\Rightarrow 2yy' - 2\tan x = y'$

At $(0, 0)$ $y' = 0$

At $(0, 1)$ $y' = 0$

Differentiating wrt x

$2yy'' + 2(y')^2 - 2\sec^2 x = y''$

At $(0, 0)$ $y'' = -2$

At $(0, 1)$ $y'' = 2$

$\therefore \quad |y''(0)| = 2$

Q.20 For the frequency distribution :

Variate (x) : x_1 x_2 $x_3 \ldots x_{15}$

Frequency (f) : f_1 f_2 $f_3 \ldots f_{15}$

where $0 < x_1 < x_2 < x_3 < \ldots < x_{15} = 10$ and $\sum_{t=1}^{15} f_t > 0$, the standard deviation cannot be

(1) 1 (2) 4 (3) 2 (4) 6

Ans. **[4]**

Sol. $\because \sigma^2 \le \frac{1}{4}(M - m)^2$

Where M and m are upper and lower bounds of values of any random variable.

$\because \sigma^2 \le \frac{1}{4}(10-0)^2$

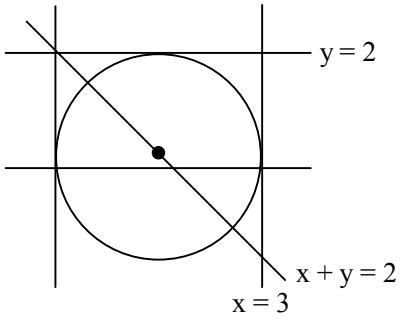
$\Rightarrow 0 < \sigma < 5$

$\therefore \sigma \ne 6.$

Q.21 The diameter of the circle, whose centre lies on the line $x + y = 2$ in the first quadrant and which touches both the lines $x = 3$ and $y = 2$, is _____ .

Ans. **[3]**

Sol.



$\because$ center lies on $x + y = 2$ and in 1st quadrant

center $= (\alpha, 2-\alpha)$

where $\alpha > 0$ and $2-\alpha > 0 \Rightarrow 0 < \alpha < 2$

$\because$ circle touches $x = 3$ and $y = 2$

$\Rightarrow |3-\alpha| = |2-(2-\alpha)| =$ radius

$\Rightarrow |3-\alpha| = |\alpha| \Rightarrow \alpha = \frac{3}{2}$

$\therefore$ radius $= \alpha$

$\Rightarrow$ Diameter $= 2\alpha = 3$

Q.22 If $\lim\limits_{x\to 0}\left\{\frac{1}{x^8}\left(1-\cos\frac{x^2}{2}-\cos\frac{x^2}{4}+\cos\frac{x^2}{2}\cos\frac{x^2}{4}\right)\right\} = 2^{-k}$, then the value of k is _____ .

Ans. **[8]**

Sol. $\lim\limits_{x\to 0}\left\{\frac{1}{x^8}\left(1-\cos\frac{x^2}{2}-\cos\frac{x^2}{4}+\cos\frac{x^2}{2}\cos\frac{x^2}{4}\right)\right\} = 2^{-k}$

$$\Rightarrow \lim_{x\to 0}\frac{\left(1-\cos\frac{x^2}{2}\right)}{4\left(\frac{x^2}{2}\right)^2}\,\frac{\left(1-\cos\frac{x^2}{4}\right)}{16\left(\frac{x^2}{4}\right)^2} = \frac{1}{8}\times\frac{1}{32} = 2^{-k}$$

$\Rightarrow 2^{-8} = 2^{-k} \Rightarrow k = 8.$

Q.23 Let $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$, $x \in R$ and $A^4 = [a_{ij}]$. If $a_{11} = 109$, then a_{22} is equal to _____ .

Ans. **[10]**

Sol. $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$

$A^2 = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$

$A^4 = \begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}\begin{bmatrix} x^2+1 & x \\ x & 1 \end{bmatrix}$

$= \begin{bmatrix} (x^2+1)^2+x^2 & x(x^2+1)+x \\ x(x^2+1)+x & x^2+1 \end{bmatrix}$

$a_{11} = (x^2+1)^2 + x^2 = 109$

$\Rightarrow x = \pm 3$

$a_{22} = x^2 + 1 = 10$

Q.24 The value of $(0.16)^{\log_{2.5}\left(\frac{1}{3}+\frac{1}{3^2}+\frac{1}{3^3}+.....\text{to }\infty\right)}$ is equal to _____.

(1) (2) (3) (4)

Ans. **[4]**

Sol. $(0.16)^{\log_{2.5}\left(\frac{1}{3}+\frac{1}{3^2}+..... \text{ to }\infty\right)}$

$= \left(\frac{4}{25}\right)^{\log_{\left(\frac{5}{2}\right)}\left(\frac{1}{2}\right)}$

$= \left(\frac{1}{2}\right)^{\log_{\left(\frac{5}{2}\right)}\left(\frac{4}{25}\right)} = \left(\frac{1}{2}\right)^{-2} = 4$

Q.25 If $\left(\frac{1+i}{1-i}\right)^{m/2} = \left(\frac{1+i}{1-i}\right)^{n/3} = 1$, $(m, n \in N)$, then the greatest common divisor of the least values of m and n is _____.

(1) (2) (3) (4)

Ans. **[4]**

Sol. $\left(\frac{1+i}{1-i}\right)^{m/2} = \left(\frac{1+i}{i-1}\right)^{n/3} = 1$

$\Rightarrow \left(\frac{(1+i)^2}{2}\right)^{m/2} = \left(\frac{(1+i)^2}{-2}\right)^{n/3} = 1$

$\Rightarrow (i)^{m/2} = (-i)^{n/3} = 1$

$\Rightarrow \frac{m}{2} = 4k_1$ and $\frac{n}{3} = 4k_2$

$\Rightarrow m = 8k_1$ and $n = 12\ k_2$

Least value of $m = 8$ and $n = 12$.

$\therefore$ GCD $= 4$